

DATA ANALYTICS PROJECT REPORT

Customer Shopping Behavior Analysis

Uncovering Spending Patterns, Customer Segments, and Loyalty Drivers Across 3,900 Retail Transactions

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Tools Used: Python (pandas) · PostgreSQL · Power BI

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1. Executive Summary

This report documents an end-to-end analysis of 3,900 retail transactions to uncover the behavioral drivers behind customer spending, loyalty, and product preference. The project followed a full analytics pipeline: raw data was cleaned and feature-engineered in Python, loaded into a PostgreSQL database to simulate live business transactions, queried with SQL to answer ten targeted business questions, and finally visualized in an interactive Power BI dashboard for stakeholder consumption.

The analysis shows a customer base that is broad-based and fairly loyal (3,116 of 3,900 customers, or roughly 80%, qualify as "Loyal"), but one where subscription conversion (27%) lags far behind engagement, and where several products depend heavily on discounting to sell. Male customers currently generate more than double the revenue of female customers, and revenue is fairly evenly spread across age groups, with Young Adults contributing the most. These patterns translate into five concrete recommendations covering subscription growth, loyalty rewards, discount-policy review, product positioning, and targeted marketing, detailed in Section 7.

2. Problem Statement & Business Objective

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has observed shifts in purchasing patterns across demographics, product categories, and channels, and wants to know which factors — discounts, reviews, seasonality, or payment preferences — most influence purchase decisions and repeat behavior.

Guiding business question:

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize product strategy?”

Project deliverables:

- **Data preparation & modeling (Python):** clean and transform the raw dataset for analysis.
- **Data analysis (SQL):** structure the data, simulate business transactions, and query it to extract insight on segments, loyalty, and purchase drivers.
- **Visualization & insights (Power BI):** build an interactive dashboard for stakeholder decision-making.
- **Report and presentation:** summarize findings and recommendations for both technical and non-technical audiences.
- **GitHub repository:** package the Python scripts, SQL queries, and dashboard file in a well-structured repo.

3. Dataset Overview

The raw dataset (Customer_Purchase_Data.csv) contains 3,900 rows and 18 columns spanning the full shopping journey — demographics, purchase details, and behavioral signals.

Metric	Value
Total transactions	3,900
Total columns	18
Unique customers	3,900 (1 transaction per customer ID)
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Missing values	37 (Review Rating column only)
Gender split	Male 2,652 · Female 1,248 (approx., Male is the modal category)
Subscription rate	27% (1,053 of 3,900)

Table 1 — High-level dataset profile

Column groups:

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method

4. Methodology

4.1 Data Preparation & Cleaning (Python)

The dataset was loaded and cleaned in pandas following a repeatable, auditable process:

- **Initial exploration:** used `df.info()` and `df.describe(include='all')` to profile structure, types, and summary statistics before making any changes.
- **Missing data handling:** 37 nulls in Review Rating were imputed using the median rating within each product Category (rather than a single global median), preserving category-level rating tendencies.
- **Column standardization:** all column names were lower-cased and converted to snake_case (e.g. "Purchase Amount (USD)" → `purchase_amount`) for consistent, error-resistant referencing in SQL and code.
- **Feature engineering — age_group:** customers were binned into four quartile-based age groups (Young Adults, Adults, Middle-aged, Senior) using `pd.qcut`, ensuring roughly equal-sized cohorts rather than arbitrary fixed-width bins.
- **Feature engineering — purchase_frequency_days:** the categorical `frequency_of_purchases` field (e.g. "Weekly", "Fortnightly", "Annually") was mapped to a numeric day count, making cadence usable in quantitative analysis.
- **Redundancy check:** `discount_applied` and `promo_code_used` were verified to be identical across all 3,900 rows; `promo_code_used` was dropped to avoid double-counting the same signal.

- **Database integration:** the cleaned DataFrame was loaded into a PostgreSQL database (via SQLAlchemy + psycopg2) as a customer table, simulating a real business transactions store for downstream SQL analysis.

4.2 Data Analysis (SQL / PostgreSQL)

With clean data loaded into PostgreSQL, ten business questions were translated into SQL queries using aggregation (GROUP BY, SUM, AVG, COUNT), filtering, conditional logic (CASE WHEN) for segmentation, and window functions (RANK/ROW_NUMBER) for top-N-per-category queries. Full results are presented in Section 5.

4.3 Visualization (Power BI)

The query outputs were consolidated into a single interactive Power BI dashboard, giving stakeholders self-serve access to the same insights through KPI cards, a subscription donut chart, and revenue/sales bar charts by Category and Age Group — filterable by Subscription Status, Gender, Category, and Shipping Type. See Section 6.

5. Key Findings — Business Questions & Results

Each of the ten business questions defined in the problem statement was answered with a dedicated SQL query. Results and interpretation follow below.

Q1. What is the total revenue generated by male vs. female customers?

Gender	Total Revenue (USD)
Female	\$75,191
Male	\$157,890

Insight: Male customers drive the large majority of revenue — \$157,890 vs. \$75,191 for female customers, a more than 2:1 split. Given men also outnumber women in the customer base (2,652 vs. 1,248), this reflects both a larger male segment and consistent average spend, pointing to a male-dominated customer base and an opportunity to grow the smaller, currently under-monetized female segment.

Q2. Which customers used a discount but still spent more than the average purchase amount?

customer_id (sample)	purchase_amount (USD)
2	\$64
3	\$73
4	\$90
7	\$85
9	\$97
12	\$68

Insight: 839 customers used a discount yet still spent above the \$59.76 average — about 22% of the customer base. This is a valuable segment: they are price-sensitive enough to use a discount but not purely discount-driven, since

their basket value remains above average. They are strong candidates for upsell and loyalty offers rather than deeper discounting.

Q3. Which are the top 5 products with the highest average review ratings?

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Ratings cluster tightly near the dataset average (3.75), with the top 5 products all sitting within 0.11 points of it. No single product stands out as a clear customer favorite, suggesting overall satisfaction is fairly uniform across the catalog rather than concentrated in a few hero products.

Q4. Compare the average purchase amount between Standard and Express shipping.

Shipping Type	Avg. Purchase Amount (USD)
Standard	\$58.46
Express	\$60.48

Insight: Express-shipping customers spend marginally more (\$60.48 vs. \$58.46, a ~3.5% difference). The gap is real but modest — express shipping is associated with slightly higher-value baskets, but it is not, on its own, a strong lever for driving up average order value.

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

Subscription Status	Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Insight: Average spend is virtually identical between subscribers (\$59.49) and non-subscribers (\$59.87) — subscription status has no meaningful effect on basket size. However, non-subscribers make up 73% of customers and therefore generate 73% of total revenue (\$170,436). Subscription today functions as a retention/engagement label rather than a spend driver, and the large non-subscriber base represents the biggest untapped opportunity for conversion campaigns.

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

Rank	Product	Discount Rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%

Rank	Product	Discount Rate
5	Pants	47.37%

Insight: Roughly half of all purchases of these five items involve a discount, well above the dataset-wide baseline (discount usage is applied to a minority of overall transactions). Hats, in particular, sell at a discount exactly half the time. These products may be over-reliant on promotions to move volume, which is worth investigating from a margin perspective — notably, Hat also appears in the top-5-by-rating list (Q3), so its discount dependency isn't a quality issue.

Q7. Segment customers into New, Returning, and Loyal based on total previous purchases, and show the count of each segment.

Segment	Number of Customers	Share of Base
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Insight: The customer base is overwhelmingly Loyal (nearly 80%), with very few New customers (2.1%). This is a mature, retention-heavy base rather than one that is actively growing through fresh acquisition — reinforcing the case for loyalty-program investment (to protect the 80% majority) alongside renewed top-of-funnel acquisition efforts (to grow the thin 2% New segment).

Q8. What are the top 3 most purchased products within each category?

Category	#1 Product (orders)	#2 Product (orders)	#3 Product (orders)
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

Insight: Order volumes are remarkably even within each category (e.g. Clothing's top three all sit within 2 orders of each other), indicating broad, balanced demand rather than one breakout bestseller per category. This makes assortment-wide merchandising strategies more effective than single-SKU promotion.

Q9. Are repeat buyers (more than 5 previous purchases) also more likely to subscribe?

Subscription Status	Repeat Buyers (>5 previous purchases)
No	2,518
Yes	958

Insight: No — repeat buying and subscribing are largely independent behaviors. Among repeat buyers, only $958 / (958 + 2,518) \approx 27.6\%$ are subscribers — essentially identical to the overall subscription rate of 27% across the whole customer base. Being a frequent purchaser does not, on its own, predict subscription; the subscription program is not currently capturing the loyalty of its best customers, which reinforces the recommendation to actively promote subscription benefits to repeat buyers rather than assuming it will happen organically.

Q10. What is the revenue contribution of each age group?

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Insight: Revenue is broadly balanced across age groups, with only an 11% spread between the highest (Young Adult, \$62,143) and lowest (Senior, \$55,763) contributors. Since the age bins were built with `pd.qcut` to hold roughly equal customer counts, this confirms per-customer spend is fairly consistent by age — the product appeals broadly across generations rather than skewing toward one age cohort.

6. Interactive Power BI Dashboard

All ten findings were consolidated into a single-page interactive dashboard, giving stakeholders self-serve access to the same insights without needing to run SQL. The dashboard includes:

- KPI cards for Number of Customers (3.9K), Average Purchase Amount (\$59.76), and Average Review Rating (3.75)
- A donut chart showing the 27% / 73% subscriber vs. non-subscriber split
- Bar charts for Revenue by Category and Sales (order count) by Category
- Bar charts for Revenue by Age Group and Sales by Age Group
- Slicers for Subscription Status, Gender, Category, and Shipping Type, enabling stakeholders to filter every visual interactively

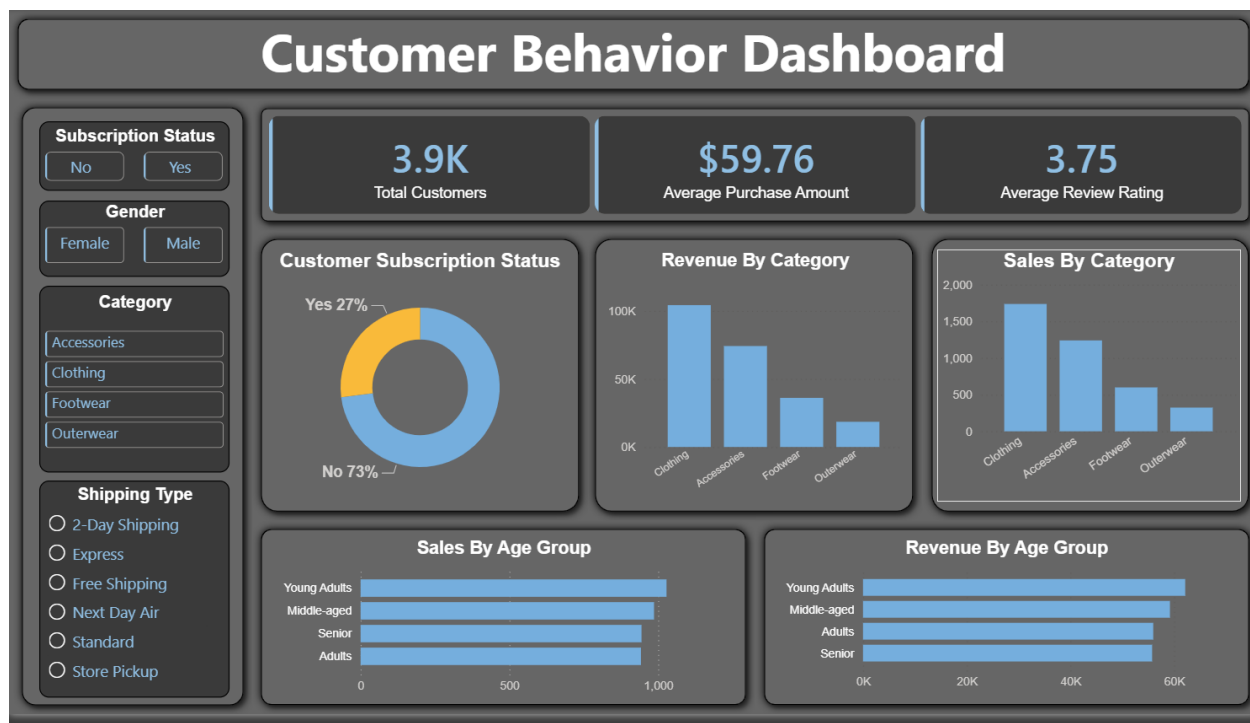


Figure 1 — Customer Behavior Dashboard (Power BI)

7. Business Recommendations

- **Boost subscriptions:** since average spend is nearly identical across subscription status, the 73% non-subscriber base (\$170,436 in revenue) is the single largest growth lever. Promote exclusive, high-visibility benefits (early access, free express shipping, subscriber-only discounts) to convert this group rather than competing purely on price.
- **Build a loyalty program for repeat buyers:** Q9 shows that being a repeat buyer does not currently increase the odds of subscribing. Introduce a structured rewards/loyalty tier targeted specifically at customers with 5+ previous purchases to convert high-frequency shoppers into subscribers and move more of the "Returning" segment (701 customers) into "Loyal."
- **Review discount policy on dependent products:** Hat, Sneakers, Coat, Sweater, and Pants each sell at a discount roughly half the time (Q6). Evaluate whether list prices, bundling, or promotional cadence for these five SKUs can be adjusted to protect margin without suppressing volume.
- **Lead marketing with top-rated and balanced-demand products:** Gloves, Sandals, Boots, Hat, and Skirt have the highest average ratings (Q3), and order volumes are evenly spread within each category (Q8) rather than concentrated in one bestseller. Feature top-rated items in campaigns and use assortment-wide (not single-SKU) merchandising.
- **Target high-revenue age groups and express-shipping users:** Young Adults and Middle-aged customers contribute the most revenue (Q10), and Express-shipping customers spend modestly more per order (Q4). Prioritize these segments in paid acquisition and retention campaigns, while continuing to serve Adult and Senior segments, whose revenue contribution is only marginally lower.
- **Grow female customer revenue:** male customers currently generate more than double the revenue of female customers (Q1). Investigate whether this reflects an imbalanced customer base, category assortment, or marketing mix, and consider targeted campaigns to grow the female segment.

8. Conclusion

This analysis moved raw transactional data through a complete, reproducible analytics pipeline — Python for cleaning and feature engineering, PostgreSQL for structured business-question querying, and Power BI for stakeholder-facing visualization. The resulting picture is of a loyal, broad-based customer population (80% "Loyal") that is under-converted on subscription (27%) and unevenly monetized by gender, with a handful of products showing heavy discount dependency. The six recommendations above give the business concrete, data-backed levers — subscription growth, loyalty rewards, discount-policy review, product positioning, targeted marketing, and gender-balanced growth — to convert these insights into improved sales, satisfaction, and long-term loyalty.

9. Appendix

9.1 Tech Stack

Layer	Tool(s)
Data preparation & feature engineering	Python (pandas)
Database / business-transaction simulation	PostgreSQL (via SQLAlchemy + psycopg2)

Layer	Tool(s)
Analysis queries	SQL (aggregation, CASE WHEN segmentation, window functions)
Visualization	Power BI (KPI cards, slicers, bar/donut charts)
Version control	GitHub repository (scripts, SQL, dashboard file)

9.2 Deliverables Checklist

- Data preparation & modeling (Python) — cleaned, feature-engineered dataset
- Data analysis (SQL) — 10 business questions answered against PostgreSQL
- Visualization & insights (Power BI) — interactive dashboard with slicers
- Report and presentation — this document + stakeholder slide deck
- GitHub repository — Python scripts, SQL queries, dashboard file

9.3 Note on Repository Hygiene

The notebook currently connects to PostgreSQL with a hardcoded username/password in-line. Before pushing the notebook to the GitHub repository called for in Deliverable 5, it's worth moving those credentials to environment variables (or a .env file excluded via .gitignore) so they aren't committed in plain text alongside the analysis.